

1. Sort an array in ascending and descending order.

```
package Array;
import java.util.Scanner;
public class Sort_array {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n=sc.nextInt();
        int[] arr=new int[n];
        System.out.println("Enter elements:");
        for (int i=0; i<n;i++)
        {
            arr[i]=sc.nextInt();
        }

        for(int i=0;i<n;i++)
        {
            for(int j=0;j<n-1;j++)
            {
                if(arr[j]>arr[j+1])
                {
                    int temp=arr[j];
                    arr[j]=arr[j+1];
                    arr[j+1]=temp;
                }
            }
        }

        for(int i=0;i<n;i++)
        {
            System.out.print(arr[i]+" ");
        }
    }
}
```

Enter size of array: 5

Enter elements:

4

3

1

2

6

1 2 3 4 6

2. Remove duplicate elements from an array.

```
package Array;
import java.util.Scanner;
public class Remove_dup {
```

```

public static void main(String[] args) {
    // TODO Auto-generated method stub
    Scanner sc=new Scanner(System.in);
    System.out.print("Enter size of array: ");
    int n=sc.nextInt();
    int[] arr=new int[n];
    System.out.println("Enter elements:");
    for (int i=0; i<n;i++)
    {
        arr[i]=sc.nextInt();
    }

    for(int i=0;i<n;i++)
    {
        for(int j=i+1;j<n; j++)
        {
            if(arr[i]==arr[j])
            {
                for(int k=j; k<n-1; k++)
                {
                    arr[k]=arr[k+1];
                }
                n--;
                j--;
            }
        }
    }

    for(int i=0;i<n;i++)
    {
        System.out.print(arr[i]+" ");
    }
}

```

Enter size of array: 5

Enter elements:

5

4

5

2

1

5 4 2 1

3. Find largest element in each row.

```

package Array;
import java.util.Scanner;
public class Largest_row_element {
    public static void main(String[] args) {

```

```

// TODO Auto-generated method stub
Scanner sc=new Scanner(System.in);
System.out.print("Enter row: ");
int row=sc.nextInt();
System.out.print("Enter column: ");
int col=sc.nextInt();
int[][] arr=new int[row][col];
System.out.println("Enter elements:");
for (int i=0; i<row;i++)
{
    for(int j=0; j<col; j++)
    {
        arr[i][j]=sc.nextInt();
    }
}

for(int i=0;i<row;i++)
{
    int max=arr[i][0];
    for(int j=1;j<col; j++)
    {
        if(arr[i][j]>max) {
            max=arr[i][j];
        }
    }
    System.out.println("Maximum element in row"+(i+1)+" : "+max);
}
}

```

Enter row: 3
Enter column: 3
Enter elements:

5
9
6
4
8
2
1
7
3

Maximum element in row1: 5
Maximum element in row2: 8
Maximum element in row3: 3

4. Check if array is palindrome.

```
package Array;
```

```

import java.util.Scanner;
public class Palindrome {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the size: ");
        int n = sc.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        boolean isPalindrome = true;
        int left = 0;
        int right = n - 1;
        while (left < right) {
            if (arr[left] != arr[right]) {
                isPalindrome = false;
                break;
            }
            left++;
            right--;
        }
        if (isPalindrome) {
            System.out.println("The array is Palindrome.");
        } else {
            System.out.println("The array is not Palindrome.");
        }
    }
}

```

Enter the size: 4

Enter elements:

5

9

4

6

The array is not Palindrome.

5. Rotate matrix by 90 degrees.

```

package Array;
import java.util.Scanner;
public class Rotate_matrix {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of matrix: ");
    }
}

```

```

int n = sc.nextInt();
int[][] arr = new int[n][n];
System.out.println("Enter arr elements:");
for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        arr[i][j] = sc.nextInt();
    }
}

for(int i=0; i<n ;i++) {
    for(int j=i;j<n;j++) {
        int temp=arr[i][j];
        arr[i][j]=arr[j][i];
        arr[j][i]=temp;
    }
}

for (int i = 0; i < n; i++) {
    int left = 0;
    int right = n - 1;
    while (left < right) {
        int temp = arr[i][left];
        arr[i][left] = arr[i][right];
        arr[i][right] = temp;
        left++;
        right--;
    }
}

System.out.println("\nAfter 90 degree rotation:");
for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        System.out.print(arr[i][j] + " ");
    }
    System.out.println();
}
}

```

Enter size of matrix: 4

Enter arr elements:

1
2
3
4
5
6
7
8

9

10

11

12

13

14

15

16

After 90 degree rotation:

13 9 5 1

14 10 6 2

15 11 7 3

16 12 8 4